

# SATHISH R

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## Education

**Rajalakshmi Engineering College, Chennai**

*B. Tech in Information Technology*

**Chennai, Tamil Nadu, India**

*Sep 2024 – Present*

*CGPA : 8.02*

## Summary

Backend Engineering undergraduate with hands-on experience developing scalable applications using Java, Spring Boot, MongoDB, and REST APIs. Built full-stack and machine learning projects involving authentication, deployment, and predictive analytics. Passionate about solving real-world problems through efficient backend systems and continuously improving software engineering skills.

## Technical Skills

**Programming** : Java, Python , C

**Web Development** : HTML, CSS, React, Spring Boot

**Database Management** : MySQL, MongoDB

**Tools** : VS Code, Jupyter Notebook, Git, GitHub

**Concepts** : Object Oriented Programming, Data Structures, Machine Learning

## Projects

### Developer Knowledge Vault

*React, Spring Boot, MongoDB*

- Developer notes and coding references were scattered across multiple platforms with no centralized system for retrieval or organization.
- Designed and built a full-stack knowledge management platform to centralize, search, and version-control developer notes.
- Engineered Spring Boot 3.2.5 backend with 17 REST endpoints, JWT authentication, MongoDB Text Indexing, hybrid React-side filtering, and automatic version history snapshots.
- Delivered a secure multi-user platform with instant zero-latency search, access analytics dashboard, and full revision history for every note.

### Crop & Carbon Prediction System

*Python, FastAPI, Scikit-learn, Docker*

- Farmers lacked accessible data-driven tools to determine suitable crops based on soil and climate conditions.
- Built an end-to-end ML web application predicting optimal crops and soil carbon sequestration potential.
- Trained Random Forest Classifier and Regressor on 7 parameters (N, P, K, pH, rainfall, temperature, humidity), served via FastAPI, containerized with Docker.
- Deployed live on Render cloud with real-time dual-model predictions and a publicly accessible web interface.
- Website link : <https://crop-and-carbon-score-prediction.onrender.com>

## Achievements

- Solved 90+ Data Structures and Algorithms problems on LeetCode, covering arrays, linked lists, stacks, queues and recursion.
- Earned the 50 Days LeetCode Badge through consistent daily problem-solving.

## Certifications

- Artificial Intelligence & Machine Learning – IITM Pravartak
- Software Engineering Virtual Experience – JPMorgan Chase & Co. (Forage).